

Augmenting fields by ordered abelian groups

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Valued fields

Definition (Valued fields)

A *valued field* is a field K , along with a *valuation* v , a homomorphism onto an ordered abelian group $v : K^\times \rightarrow \Gamma$ such that $v(x + y) \geq \min(v(x), v(y))$, with the convention that $v(0) = \infty$. We call $\Gamma = vK$ the *value group*.

The *valuation ring* is $\mathcal{O}_v = \{x \in K \mid v(x) \geq 0\}$, and its only maximal ideal is $\mathfrak{m}_v = \{x \in K \mid v(x) > 0\}$. We call $Kv = \mathcal{O}_v/\mathfrak{m}_v$ the *residue field*.

We will work on the languages $\mathcal{L}_{\text{ring}} = \{+, -, \cdot, 0, 1\}$ and $\mathcal{L}_{\text{val}} = \mathcal{L}_{\text{ring}} \cup \{\mathcal{O}\}$.

Remark

We can interpret all the objects above in the $\mathcal{L}_{\text{val}}$ -structure $(K, \mathcal{O}_v)$:

$$(\Gamma, \Gamma_{\geq 0}, +) \cong (K^\times/\mathcal{O}_v^\times, \mathcal{O}_v^*/\mathcal{O}_v^\times, \cdot), \quad \mathfrak{m}_v = \mathcal{O}_v \setminus \mathcal{O}_v^\times, \quad Kv = \mathcal{O}_v/\mathfrak{m}_v$$

We say the valuation v is *definable* if $\mathcal{O}_v$ is $\mathcal{L}_{\text{ring}}$ -definable in K .

Valued fields

Definition (Henselianity)

(K, v) is *henselian* if the following, equivalent conditions hold:

- All simple roots of $P(x) \in \mathcal{O}_v[x]$ in Kv lift to a root in $\mathcal{O}_v$.
- There is only one extension of v from K to K^{alg} .

Theorem (Ax-Kochen, Ershov; 1965)

Let $(K, v), (L, w)$ be henselian of residue characteristic zero. Then

$$(K, v) \equiv (L, w)$$

if and only if

$$vK \equiv wL \quad \text{and} \quad Kv \equiv Lw.$$

The theorem still holds if we change “ $\equiv$ ” by “ $\preceq$ ”.

Valued fields

Let $\mathbf{k}$ be a field and Γ an ordered abelian group (oag).

Definition (Hahn series)

A formal power series

$$f(t) = \sum_{\gamma \in \Gamma} a_{\gamma} t^{\gamma}$$

with $(a_{\gamma})_{\gamma \in \Gamma} \subseteq \mathbf{k}$ is a *Hahn series* if $\text{supp}(f)$ is well-ordered.
Denote by the field of all Hahn series by $\mathbf{k}((t^{\Gamma}))$ or $\mathbf{k}((\Gamma))$.

The field $K = \mathbf{k}((\Gamma))$ admits a henselian valuation $v = v_{\Gamma}$ given by $v(f) = \min \text{supp}(f)$, such that $Kv = \mathbf{k}$ and $vK = \Gamma$.

Fact

Let $\mathbf{k}$ be a field and let Γ_1, Γ_2 be oags. Then

$$\mathbf{k}((\Gamma_1))((\Gamma_2)) = \mathbf{k}((\Gamma_2 \oplus \Gamma_1)).$$

Henselianity in $\mathcal{L}_{\text{ring}}$

Proposition (Robinson; 1949)

Let (K, v) be a henselian valued field with $vK = \mathbb{Z}$ and $\text{char}(Kv) \neq 2$. Let π be a uniformizer. Then $\mathcal{O}_v$ is defined by

$$\exists y : 1 + \pi x^2 = y^2.$$

If $\text{char}(Kv) = 2$, we can use $\exists y : 1 + \pi^3 x^4 = y^2$ instead.

The ring $\mathbb{Z}_p$ is $\emptyset$ -defined in $\mathbb{Q}_p$ by $\exists y : 1 + px^2 = y^2$. The theory $\text{Th}(\mathbb{Q}_p, v_p)$ reduces to $\text{Th}(\mathbb{Q}_p)$.

Similarly, $\mathbb{C}[[t]]$ is defined in $\mathbb{C}((t))$ by $\exists y : 1 + tx^2 = y^2$, using only the parameter t .

There is an analogous formula whenever $vK = \Gamma$ has a minimum positive element.

Henselianity in $\mathcal{L}_{\text{ring}}$

There are nice cases where the valuations are not definable.

Example ($\mathbf{ACF}_0$)

Any field $K \equiv \mathbb{C}$ admits a non-trivial henselian valuation, as $K \models \mathbf{ACF}_0$. However, none of these valuations are definable by strong minimality.

Example ($\mathbf{RCF}$)

Let $R \equiv \mathbb{R}$. By Schmidt's Theorem, any valuation on R is henselian iff it is convex. By o-minimality, a convex, definable set must be an interval and thus it cannot be a non-trivial subring. Thus, no non-trivial henselian valuation in R is definable.

Furthermore, $\mathbb{R}$ has no non-trivial convex subrings, so it admits no henselian valuation. However, if ${}^*\mathbb{R} \succeq \mathbb{R}$, the ring $\mathcal{O}$ of finite numbers is a henselian valuation ring.

Henselianity in $\mathcal{L}_{\text{ring}}$

Let K and $\mathbf{k}$ be fields.

Definition

- K is *henselian* if it admits a non-trivial henselian valuation.
- K is $(\emptyset)$ -definably henselian if it admits a non-trivial $(\emptyset)$ -definable henselian valuation.
- K is *elementarily henselian* if any $\mathbf{k} \equiv K$ is henselian.
- $\mathbf{k}$ is *t-henselian* if there is a henselian $K \equiv \mathbf{k}$.
- $\mathbb{Q}$ is not *t-henselian*.
- $\mathbb{R}$ is *t-henselian*, but not henselian.
- ${}^*\mathbb{R}$ is henselian, but not elementarily henselian.
- $\mathbb{C}$ is elementarily henselian, but not definably henselian.
- $\mathbb{Q}_p$ is $\emptyset$ -definably henselian and thus elementarily henselian.

Henselianity in $\mathcal{L}_{\text{ring}}$

Example (Boissonneau, de Mase, Jahnke, Touchard; 2025)

Let $\Gamma = \Delta \oplus \Delta \oplus \Delta \oplus \cdots$ and $G = \cdots \oplus \Delta \oplus \Delta \oplus \Delta$, with lexicographic order. Notice that $\Gamma \equiv \Delta \oplus \Gamma$ and $G \equiv G \oplus \Delta$.

If $\mathbf{k} = \mathbb{C}((\Gamma))$, notice that

$$\mathbf{k} = \mathbb{C}((\Gamma)) \equiv \mathbb{C}((\Delta \oplus \Gamma)) = \mathbb{C}((\Gamma))((\Delta)) = \mathbf{k}((\Delta))$$

and thus $\mathbf{k} \equiv \mathbf{k}((\Delta))$ by AKE.

Let $(K, v) = (\mathbf{k}((G)), v_G)$, and consider the valuations on L/K given by

$$L = \mathbf{k}((\Delta)) \underbrace{((G))}_{w_1} = \mathbf{k} \underbrace{((G \oplus \Delta))}_{w_2}.$$

By AKE, we have $(K, v) \equiv (L, w_1) \equiv (L, w_2)$, so (K, v) can not be $\emptyset$ -definable.

If $\Delta = \mathbb{Z}$, then $G = \cdots \oplus \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}$ has a minimum element $(\dots, 0, 0, 1)$, thus (K, v) is definable by Robinson's formula.

Augmentable fields

Let $\mathbf{k}$ be a field, Γ and Δ be oags. From now on, we always assume $\text{char}(\mathbf{k}) = 0$.

Definition

$\mathbf{k}$ is (strongly) Δ -augmentable if $\mathbf{k} \preceq \mathbf{k}((\Delta))$.

Γ is (strongly) Δ -augmentable if $\Gamma \preceq \Delta \oplus \Gamma$.

The weak version is obtained by changing “ $\preceq$ ” by “ $\equiv$ ”.

Notice that weak Δ -augmentability on a field implies t -henselianity.

Being $\mathbb{Q}$ -augmentable seems like a common condition:

Example

As $\mathbb{C}((\mathbb{Q})) \models \mathbf{ACF}$, $\mathbb{R}((\mathbb{Q})) \models \mathbf{RCF}$ and $\mathbb{Q}_p((\mathbb{Q})) \models \mathbf{pCF}$ and these theories are model-complete, we have $\mathbb{C} \preceq \mathbb{C}((\mathbb{Q}))$, $\mathbb{R} \preceq \mathbb{R}((\mathbb{Q}))$ and $\mathbb{Q}_p \preceq \mathbb{Q}_p((\mathbb{Q}))$.

Thus $\mathbb{C}, \mathbb{R}$ and $\mathbb{Q}_p$ are $\mathbb{Q}$ -augmentable.

In private communication, van den Dries suggested $\mathbb{Q}$ -augmentability should correspond to t -henselianity, at least in characteristic zero.

Augmentable fields

Not all t -henselian fields are $\mathbb{Q}$ -augmentable.

Example

Let $\Gamma = \cdots \oplus \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}$ and $\mathbf{k} = \mathbb{C}((\Gamma))$, which is henselian. Suppose $\mathbf{k}$ is $\mathbb{Q}$ -augmentable. Then, we would have

$$\mathbb{C}((\Gamma)) = \mathbf{k} \preceq \mathbf{k}((\mathbb{Q})) = \mathbb{C}((\Gamma))((\mathbb{Q})) = \mathbb{C}((\mathbb{Q} \oplus \Gamma))$$

as rings. However, Γ and $\mathbb{Q} \oplus \Gamma$ have the same minimum element $(\dots, 0, 0, 1)$, so the valuations are definable with the same parameter and thus $\mathbb{C}((\Gamma)) \preceq \mathbb{C}((\mathbb{Q} \oplus \Gamma))$ as valued fields.

It follows that $\Gamma \preceq \mathbb{Q} \oplus \Gamma$ as oags but, by the machinery of spines in [CH11], this cannot be the case.

However, we have

Fact ([BDMJT25])

Any t -henselian field $\mathbf{k}$ is Δ -augmentable for some non-trivial oag Δ .

Augmentable fields

Proposition (van den Dries, Folklore)

The class of (weakly/strongly) Δ -augmentable fields of characteristic zero is elementary.

Proof.

Let $\mathbf{k} = \prod_{\mathcal{U}} \mathbf{k}_{\alpha}$. If $\mathbf{k}_{\alpha} \preceq \mathbf{k}_{\alpha}((\Delta))$ for all α , then $\mathbf{k} \preceq \prod_{\mathcal{U}} \mathbf{k}_{\alpha}((\Delta))$. By AKE $\mathbf{k}((\Delta)) \preceq \prod_{\mathcal{U}} \mathbf{k}_{\alpha}((\Delta))$, so $\mathbf{k} \preceq \mathbf{k}((\Delta))$. Thus the class is closed under ultraproducts.

Now, let $\mathbf{k} \equiv \ell$ and assume $\mathbf{k} \preceq \mathbf{k}((\Delta))$. By Keisler-Shelah, we have $\ell \preceq \mathbf{k}^{\mathcal{U}}$ for some ultrafilter $\mathcal{U}$. As Δ -augmentability is preserved by ultraproducts, we have $\mathbf{k}^{\mathcal{U}} \preceq \mathbf{k}^{\mathcal{U}}((\Delta))$. Then:

$$\begin{aligned} \ell &\subseteq \ell((\Delta)) \preceq \mathbf{k}^{\mathcal{U}}((\Delta)) \\ \ell &\preceq \mathbf{k}^{\mathcal{U}} \preceq \mathbf{k}^{\mathcal{U}}((\Delta)) \end{aligned}$$

by AKE, so $\ell \preceq \ell((\Delta))$. ■

Augmentable fields

Proposition

$\mathbf{k}$ is $\mathbb{Q}$ -augmentable iff $\mathbf{k}$ is divisibly t -henselian.

Proof.

Let (K, v) be henselian with $K \equiv \mathbf{k}$ and $\Gamma = vK$ divisible. We can saturate (K, v) and coarsen v , so without loss of generality we may assume $\text{char}(Kv) = 0$. Let $(L, w) = (Kv((\Gamma)), v_\Gamma) \equiv (K, v)$, we need to prove L is $\mathbb{Q}$ -augmentable. As Γ is divisible, it is $\mathbb{Q}$ -augmentable. Thus $\Gamma \preceq \mathbb{Q} \oplus \Gamma$ and

$$L = Kv((\Gamma)) \preceq Kv((\mathbb{Q} \oplus \Gamma)) = Kv((\Gamma))((\mathbb{Q})) = L((\mathbb{Q}))$$

by AKE, so $L \equiv K \equiv \mathbf{k}$ are all $\mathbb{Q}$ -augmentable. ■

Question

Can we change “divisibly t -henselian” by “ $\mathbb{Q}$ -augmentably t -henselian”?

Augmentable groups

Fact ([BDMJT25])

Suppose $\text{char}(\mathbf{k}) = 0$. Then $\mathbf{k}$ is not t -henselian iff every henselian (K, v) with $Kv = \mathbf{k}$ is $\emptyset$ -definable.

Suppose $\mathbf{k}$ is not t -henselian. For example, if $\mathbf{k} \equiv \mathbb{Q}$, or if $\mathbf{k}$ is unstable PAC.

Lemma

An oag Γ is Δ -augmentable iff $K = \mathbf{k}((\Gamma))$ is Δ -augmentable.

Proof.

Notice that the extension $K \subseteq K((\Delta))$ is equal to $\mathbf{k}((\Gamma)) \subseteq \mathbf{k}((\Delta \oplus \Gamma))$.

If $\Gamma \preceq \Delta \oplus \Gamma$, then $\mathbf{k}((\Gamma)) \preceq \mathbf{k}((\Delta \oplus \Gamma))$ by AKE.

If $\mathbf{k}((\Gamma)) \preceq \mathbf{k}((\Delta \oplus \Gamma))$, then $\Gamma \preceq \Delta \oplus \Gamma$ by definability. ■

The same works for weakly Δ -augmentable.

Augmentable groups

Fix $\mathbf{k}$ not t -henselian.

Corollary

The class of Δ -augmentable groups is elementary.

Proof.

$\Gamma_1 \equiv \Gamma_2$ implies $\mathbf{k}(\Gamma_1) \equiv \mathbf{k}(\Gamma_2)$ by AKE. Then Γ_1 is Δ -augmentable iff $\mathbf{k}(\Gamma_1)$ is, iff $\mathbf{k}(\Gamma_2)$ is, iff Γ_2 is. The class is closed under equivalence, similar argument for ultraproducts. ■

Corollary

If Γ_1 is Δ -augmentable, then so is $\Gamma_1 \oplus \Gamma_2$.

Proof.

$$\mathbf{k}(\Gamma_1 \oplus \Gamma_2) = \mathbf{k}(\Gamma_2)((\Gamma_1)) \preceq \mathbf{k}(\Gamma_2)((\Delta \oplus \Gamma_1)) = \mathbf{k}((\Delta \oplus \Gamma_1 \oplus \Gamma_2))$$

Augmenting fields

By model-completeness of Presburger arithmetic, we have $\mathbb{Z} \preceq \mathbb{Q} \oplus \mathbb{Z}$. It follows by induction that $\mathbb{Z}^d$ is always $\mathbb{Q}$ -augmentable.

Example

Notice how

$$\mathbb{Z} \subseteq \mathbb{Z} \oplus \mathbb{Z} \subseteq \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z} \subseteq \dots$$

a chain of $\mathbb{Q}$ -augmentable groups, but its union $\dots \oplus \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}$ isn't. The quantifier complexity is greater than $\forall\exists$.

Example

If k is not t -henselian, then

$$k((\mathbb{Z})) \subseteq k((\mathbb{Z} \oplus \mathbb{Z})) \subseteq k((\mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z})) \subseteq \dots$$

is a chain of $\mathbb{Q}$ -augmentable fields. Let (K, v) be their union, so $K \preceq k((\dots \oplus \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}))$ by AKE. As $k((\dots \oplus \mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}))$ is not $\mathbb{Q}$ -augmentable, neither is K , so this class is not $\forall\exists$.

Bibliography

- [BDMJT25] Blaise Boissonneau, Anna De Mase, Franziska Jahnke, and Pierre Touchard. Growing spines ad infinitum. *arXiv preprint arXiv:2501.10531*, 2025.
- [CH11] Raf Cluckers and Immanuel Halupczok. Quantifier elimination in ordered abelian groups. *Confluentes Mathematici*, 3(4):587–615, December 2011.